

Evaluation Sprint

Performance & Security Testing

Module 12

SESSION 1

Bridge Recall: From Expansion to Testing

1. New Feature

What new feature did your team add in Sprint 2?

2. Integration

What integration did you implement?

3. Success Metric

What metric did you use to show it works (time, gas, accuracy, etc.)?

4. Prediction

What do you predict could go wrong when it's used by many users?

TRY IT OUT (15 MIN) : Break It to Make It Better

Individually Write two ways your Sprint 2 feature could fail.	ACTION Brainstorm failures
Pair Up Turn each failure into a test case sentence. <i>"When the system should (expected). We will run steps to verify"</i>	ACTION Format as test cases

Group Select Pick one best test and write it on the board.	ACTION Share best idea
Scan & Mark Scan the board. Mark any duplicates or missing ideas (e.g.. logs leak data).	ACTION Review & consolidate

Performance Testing Lab

Instructions:

1. Pick a Function
2. Implement Timing Code
3. Run 3 Cases (repeat multiple times)

Test Result Logs

TEST CASE	STEPS (SHORT)	EXPECTED	ACTUAL (time/gas)	PASS/FAIL	NOTE
Normal Input	<i>Enter standard text...</i>	<i>< 200ms</i>		☐ Pass ☐ Fail	<i>Baseline</i>
Long Input	<i>Paste 1000+ chars...</i>	<i>Process/Error</i>		☐ Pass ☐ Fail	
Empty Input	<i>Submit blank field....</i>	<i>Handle gracefully</i>		☐ Pass ☐ Fail	
Custom Case 1				☐ Pass ☐ Fail	
Custom Case 2				☐ Pass ☐ Fail	

SESSION 2

Bridge Recall: From Expansion to Testing

1. Identify the Bottleneck:

What was your slowest test case from yesterday?

2. Measurement Method:

What metric did you use to measure it (time, gas, etc.)?

3. Root Cause Analysis:

What made that case slow or costly?

4. The Next Step:

If speed is one type of reliability, what other quality makes a system trustworthy?

TRY IT OUT (15 MIN) : Break It to Make It Better

Review Prototype: Examine your prototype from Sprint 1 and 2. Look at features, inputs, and data flows.	ACTION Analyze current build.
Identify Risks: Write down two ways someone could misuse, attack, or break it.	ACTION List potential attacks.

Share & Select: Share ideas with a seatmate. Discuss the impact of each vulnerability. Pick the most realistic security risk to test in the lab.	ACTION Choose target vulnerability.
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Security Testing Lab

Instructions:

1. Input Handling Tests (3x)
2. Observe Behavior
3. Add a Guard (Optional)

Security Test Results

TEST CASE	STEPS (SHORT)	EXPECTED	ACTUAL (time/gas)	PASS/FAIL	NOTE
Authorization	<i>Call restricted fn as non-owner...</i>	<i>Revert "Not Auth"</i>		☐ Pass ☐ Fail	<i>Access Control</i>
Invalid Input	<i>Long string/zero value...</i>	<i>Revert/Safe Handle</i>		☐ Pass ☐ Fail	<i>Edge Case</i>
Data Exposure	<i>Check logs for secrets...</i>	<i>No sensitive leaks</i>		☐ Pass ☐ Fail	<i>Privacy</i>
Custom Attack 1				☐ Pass ☐ Fail	
Custom Attack 2				☐ Pass ☐ Fail	

Wrap Up

Think Back

Surprising Result:

"Our most surprising test result was..."

Reasoning:

"Why we think it happened..."

Think Back

Next Steps:

"One change we'll try next sprint..."

Wrap Up Notes
